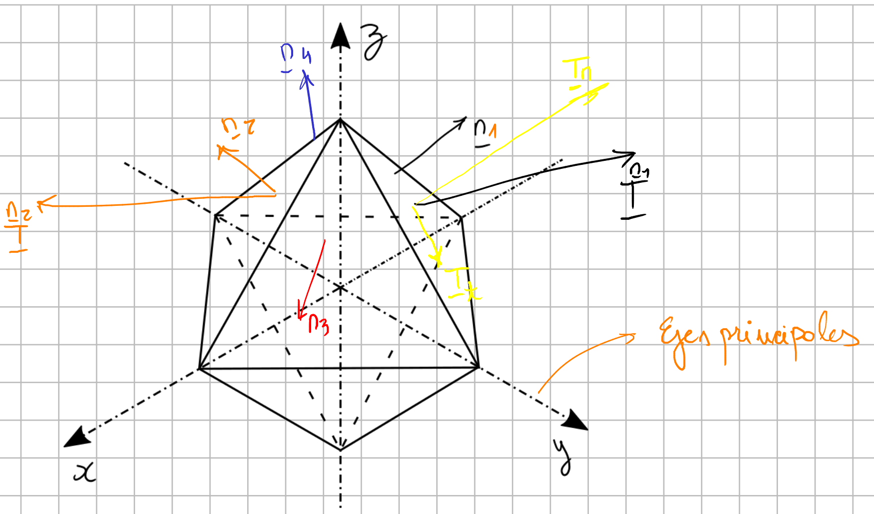



$$\underline{n}^* = \begin{pmatrix} \pm 1 \\ \pm 1 \\ \pm 1 \end{pmatrix} = \begin{pmatrix} s_1 \\ s_2 \\ s_3 \end{pmatrix}$$

$$\underline{n} = \frac{1}{\sqrt{\underbrace{s_1^2}_{1} + \underbrace{s_2^2}_{1} + \underbrace{s_3^2}_{1}}} \begin{pmatrix} s_1 \\ s_2 \\ s_3 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} s_1 \\ s_2 \\ s_3 \end{pmatrix}$$

$$\frac{1}{\sqrt{3}} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix}$$

$$\sigma_1 \geq \sigma_2 \geq \sigma_3$$

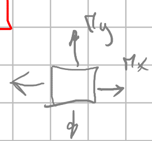
$$\underline{\underline{\sigma}} = \begin{bmatrix} \sigma_x & 0 & 0 \\ 0 & \sigma_y & 0 \\ 0 & 0 & \sigma_z \end{bmatrix} = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$$

$$\underline{\underline{\sigma}} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} \Rightarrow \sigma_M = \frac{1}{3}(\sigma_x + \sigma_y + \sigma_z)$$

$$\underline{T} = \underline{\underline{\sigma}} \cdot \underline{n} = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} \sigma_x s_1 \\ \sigma_y s_2 \\ \sigma_z s_3 \end{bmatrix}$$

$$\boxed{T_n = \underline{T} \cdot \underline{n} = \frac{1}{\sqrt{3}} \begin{bmatrix} \sigma_x s_1 \\ \sigma_y s_2 \\ \sigma_z s_3 \end{bmatrix} \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = \frac{1}{3}\left(\sigma_x \overbrace{s_1^2}^{1} + \sigma_y \overbrace{s_2^2}^{1} + \sigma_z \overbrace{s_3^2}^{1}\right) = \frac{1}{3}(\sigma_x + \sigma_y + \sigma_z) = \sigma_m}$$

Tensión media




$$\underline{\underline{\sigma}} = \underline{\underline{\sigma}}_H + \underline{\underline{\sigma}}_D$$

Tensor hidrostático $\Rightarrow \underline{\underline{\sigma}}_H = \sigma_m \cdot \underline{\underline{I}}$

Tensor desviador $\Rightarrow \underline{\underline{\sigma}}_D = \underline{\underline{\sigma}} - \underline{\underline{\sigma}}_H = \underline{\underline{\sigma}} - \sigma_m \underline{\underline{I}}$

$$\underline{T}_t = \underline{T} - T_n \cdot \underline{n} = \frac{1}{\sqrt{3}} \begin{bmatrix} \sigma_x s_1 \\ \sigma_y s_2 \\ \sigma_z s_3 \end{bmatrix} - \sigma_m \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} (\sigma_x - \sigma_m)\, s_1 \\ (\sigma_y - \sigma_m)\, s_2 \\ (\sigma_z - \sigma_m)\, s_3 \end{bmatrix}$$

$$\|\underline{T}_t\| = \frac{1}{\sqrt{3}} \sqrt{(\sigma_x - \sigma_m)^2 + (\sigma_y - \sigma_m)^2 + (\sigma_z - \sigma_m)^2}$$

hacer hasta aca por lo menos

$\hookrightarrow \sigma_m = \frac{1}{3}(\sigma_x + \sigma_y + \sigma_z)$

$$\|\underline{T}_t\| = \frac{1}{3} \sqrt{(\sigma_x - \sigma_y)^2 + (\sigma_x - \sigma_z)^2 + (\sigma_y - \sigma_z)^2}$$